

# ROHAN MATHEW S

Assistant Professor (Junior) | PhD Research Scholar, AI in Healthcare

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## PROFESSIONAL SUMMARY

Assistant Professor (Junior) and PhD research scholar at Vellore Institute of Technology, working on artificial intelligence and deep learning for early prediction of Alzheimer's disease. Background in data analytics and machine learning, with hands-on experience in Python, SQL, Tableau and explainable AI. Presented peer-reviewed research on explainable ensemble learning at ICSCSA 2026.

## EXPERIENCE

### Assistant Professor (Junior)

Jun 2026 – Present

Vellore Institute of Technology, Vellore

- Teach computer science courses while continuing doctoral research.

### PhD Research Scholar

Jul 2025 – Present

Vellore Institute of Technology, Vellore

- Research on AI and deep learning models that predict progression from mild cognitive impairment to Alzheimer's disease.
- Conducting a systematic literature review using the PICO framework and PRISMA guidelines across IEEE Xplore, PubMed, Scopus and Web of Science.
- Presented an IEEE conference paper on explainable hybrid ensemble learning for deceptive website detection at ICSCSA 2026.

### Data Analytics Intern

May 2024 – Aug 2024

GrowUp, Noida (Remote)

- Cleaned and analyzed datasets using Python and SQL to ensure data accuracy for predictive modeling.
- Designed and deployed interactive Tableau dashboards that helped stakeholders make informed decisions.
- Identified key trends and shared practical insights to improve operational efficiency.

## EDUCATION

### Doctor of Philosophy (PhD)

Jul 2025 – Present

Vellore Institute of Technology, Vellore

CGPA: 9.40 / 10

- School of Computer Science Engineering and Information Systems.
- Research area: AI and deep learning for early prediction of Alzheimer's disease.

### Master of Computer Applications (MCA)

2022 – 2024

Vellore Institute of Technology, Vellore

CGPA: 8.20 / 10

- Relevant coursework: Soft Computing, Big Data, Cloud Computing.

### Bachelor of Computer Applications (BCA)

2019 – 2022

B S Abdur Rahman University of Science and Technology, Chennai

CGPA: 8.01 / 10

- Relevant coursework: Multimedia, Web Application Development.

## PUBLICATIONS

- S. Bhuvana and Rohan Mathew S**, "Explainable Hybrid Ensemble Learning for Deceptive Website Detection with Cross-Dataset Validation," in Proceedings of the 6th International Conference on Soft Computing for Security Applications (ICSCSA-2026), IEEE, 2026. ISBN: 979-8-3315-5993-9. (Presented)

## RESEARCH INTERESTS

Deep Learning for Healthcare, Alzheimer's Disease Prediction, Explainable AI (XAI), Multimodal Learning, Machine Learning for Cybersecurity, Systematic Reviews.

## TECHNICAL SKILLS

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- **Programming:** Python, R, SQL, Java, JavaScript, C++, HTML, CSS
- **Machine Learning:** Deep Learning (CNN, RNN, LSTM), XGBoost, LightGBM, CatBoost, Random Forest, Stacking Ensembles, SHAP, Statistical Modeling
- **Data Analytics & Visualization:** Pandas, NumPy, Seaborn, Matplotlib, Plotly, Databricks, Tableau, Power BI, Excel (Advanced)
- **Databases:** MySQL, NoSQL, MongoDB, Snowflake
- **Tools:** Git, VS Code, SAS, Zotero, Agile, Scrum

## PROJECTS

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| <b>Deceptive Website Detection</b>                                                                                                                                                                                                                                                                                                                                                                                          | <b>Aug 2026</b>            |
| <ul style="list-style-type: none"><li>• Built a hybrid stacking ensemble of CatBoost, XGBoost and LightGBM to detect phishing and deceptive websites, benchmarked against conventional ML models.</li><li>• Aligned 30 URL and domain features across the Kaggle and UCI datasets and validated the model across datasets.</li><li>• Used SHAP to explain predictions; presented as an IEEE paper at ICSCSA 2026.</li></ul> |                            |
| <b>Liver Disease Prediction</b>                                                                                                                                                                                                                                                                                                                                                                                             | <b>May 2023 – Jun 2023</b> |
| <ul style="list-style-type: none"><li>• Developed an LSTM-based diagnostic system to predict liver disease.</li></ul>                                                                                                                                                                                                                                                                                                       |                            |
| <b>Diabetes Prediction</b>                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Apr 2023 – May 2023</b> |
| <ul style="list-style-type: none"><li>• Applied and compared multiple machine learning algorithms to predict diabetes.</li></ul>                                                                                                                                                                                                                                                                                            |                            |
| <b>Prescription Recognition</b>                                                                                                                                                                                                                                                                                                                                                                                             | <b>Oct 2022 – Nov 2022</b> |
| <ul style="list-style-type: none"><li>• Automated prescription reading using a CNN, RNN and LSTM pipeline.</li></ul>                                                                                                                                                                                                                                                                                                        |                            |
| <b>Fitbit Consumer Behavior Analysis Dashboard</b>                                                                                                                                                                                                                                                                                                                                                                          | <b>Aug 2024</b>            |
| <ul style="list-style-type: none"><li>• Cleaned and explored Fitbit datasets and built a Tableau dashboard of behavior patterns to support marketing strategy.</li></ul>                                                                                                                                                                                                                                                    |                            |
| <b>Indian Cars: Data Analysis and Visualization</b>                                                                                                                                                                                                                                                                                                                                                                         | <b>Jul 2024</b>            |
| <ul style="list-style-type: none"><li>• Analyzed car sales data with SQL and Python and visualized insights using histograms, bar charts, scatter plots and heat maps.</li></ul>                                                                                                                                                                                                                                            |                            |
| <b>World Bank Data Visualization</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Jun 2024</b>            |
| <ul style="list-style-type: none"><li>• Analyzed global fertility and life expectancy data (1960–2016) and built animated charts with Plotly.</li></ul>                                                                                                                                                                                                                                                                     |                            |

## CERTIFICATIONS

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- Google Analytics Certification – Google
- AWS Cloud Technical Essentials – Coursera
- Databases and SQL for Data Science with Python – IBM
- Introduction to Data Analysis – IBM
- AI for All: From Basics to GenAI Practice – NVIDIA
- Problem Solving (Basic) – HackerRank
- Python Programming, Java Programming, Basic Coding – Preplnsta